

SUBJECT	DATE	TOTAL MARKS	DURATION
PROJECT MANAGEMENT III (PRM370/PRM371S/ PRM372S)	7 OCT 2020	100	3 HOURS



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS AND DESIGN

COURSE: DIPLOMA: INFORMATION & COMMUNICATION TECHNOLOGY [DPICTA/C/M]
OFFERING: FULL-TIME AND PART-TIME

EXAMINER (S) : MR L SMALL
MODERATOR (INTERNAL) : DR E RUHODE
MODERATOR (EXTERNAL) :

SPECIAL INSTRUCTIONS

- Answer all the questions.
- This is an open book test.
- Answer all questions in the answer book, except where noted otherwise.
- Read the questions from beginning to end before attempting any question.
- When completed, scan all answers using Adobe Scan, and submit.
- WRITE YOUR NAME AND STUDENT NUMBER ON EACH PAGE.

REQUIREMENTS

- Answer book.
- Calculator.
- Adobe Scan.

PROJECT SCENARIO

Your company, UBUNTU Office Furnishings (Pty) Ltd has a large number of sales representatives who sell office furniture to major corporations. Each sales representative is assigned to a specific province, and each province is part of one of two regions in the country - inland and coastal.

To enable management to monitor the number and amount of sales for each representative, for each province, and for each region, ABC has decided to build a web-based information system that will track prices, inventory, and the competition.

The IT Department within the corporation has assigned Zandile Khumalo to be the project manager of the Web-based Reporting System development project.

Before they started the actual software development, some design decisions had to be made.

Decision Time – Managing a Risk

As part of the software development, the project had to provide the ability to view and manipulate large tables of sales data. The requirements included the ability to sort the various data columns and also to page through the various pages of data, displaying only a subset on each page.

To accomplish this was a technical challenge for the team. One of the team members suggested using a third party software control – DevExpress – to perform these tasks. Since the team had limited experience in working with these controls this was viewed as a significant project risk.

A preliminary NPV analysis was done, but further risk analysis had to be performed to come to a final decision. After some research, the team started to quantify the associated risk as follows: The options were to 1) develop the solution in-house or, 2) to purchase a third-party control;

When considering the make or buy option for implementation, you calculate that if you decide to make the software in-house, you face the following potential risk:

To develop the software in-house (Make):

- Staff may leave and need to be replaced. The consensus opinion is that there is a 50% chance of keeping the required level of staff, (and hence a 50% chance of losing key staff).
- The benefit to the project when replacing staff key staff would be R200,000. The benefit to the project when keeping key staff is R500,000.

To purchase the software (Buy), the following factors influence the project risk:

- There will be a 40% chance of requiring extensive customisation. (Hence a 60% chance of minimal customisation.) The software implementation with minimal customisation will bring in an additional R500,000.
- However, additional customisation could bring in a total of R300,000 for this option.

A decision has to be made and Zandile and her team are using the Expected Monetary Value technique to assist.

Monitoring Progress

With the help of her staff, Zandile identified all of the major tasks that need to be accomplished. The major identified tasks are Systems Design, Software Build, Systems Testing and Implementation.

The total budgeted cost (TBC) is R1,000,000. The time-phased budget is represented in the table below: Please note that all values are in thousands of rands (e.g., an amount of R4, represents R4,000 for the project).

The time phased budget breakdown is represented in the table below:

Cumulative Budgeted Cost	Week												
	TBC	1	2	3	4	5	6	7	8	9	10	11	12
Design	100	20	30	30	20								
Build	620				100	120	150	200	50				
Testing	220								80	80	40	20	
Implementation	60											35	25
CBC	1,000	20	50	80	200	320	470	670	800	880	920	975	1,000

Since the project has now been running for nine weeks, it is possible to track what has actually occurred in the project. For cost management and reporting, we can track the actual money spent on the project by getting these totals directly from the finance department.

For the first nine weeks of the project, the accounting department has provided the following detailed information:

Cumulative Actual Cost	Week												
	TAC	1	2	3	4	5	6	7	8	9	10	11	12
Design	130	40	40	30	15	5							
Build	610				30	30	200	200	150				
Testing	200								120	80			
Implementation	0												
CAC	940	40	80	110	155	190	390	590	860	940			

Zandile has also received feedback from her various team leaders regarding the activities that have been completed, and has a weekly summary of the completion of each task. The weekly task completion is indicated in the table below:

Activity Percentage Completion	Week												
		1	2	3	4	5	6	7	8	9	10	11	12
Design		10%	15%	40%	60%	100%	100%	100%	100%	100%			
Build					20%	35%	50%	80%	95%	100%			
Testing									30%	70%			
Implementation													

These reports enable Zandile to complete the earned value for each phase, and also the cumulative earned value(CEV).

QUESTION 1 – EARNED VALUE MANAGEMENT**32 MARKS**

- 1.1 Read the scenario, and then complete the table of earned value of the project for the first nine weeks in the space provided in the answer sheet.

Remember that value is only earned when it is reported that a certain percentage of the task is completed. (Please enter the values in thousands, i.e. If the value is R24,000, you will enter a value of 24) (24)

- 1.2 What is the value of EVM to project management? What potential drawbacks to heavy reliance on EVM can you envisage? (8)

QUESTION 2 – EVM ANALYSIS**24 MARKS**

The key indicators of Earned Value Management are:

- Cost Variance ($CV = CEV - CAC$)
- Schedule Variance ($SV = CEV - CBC$)
- Cost Performance Index ($CPI = CEV/CAC$)
- Schedule Performance Index ($SPI = CEV/CBC$)

Interpret the financial data to report on the following:

- 2.1 After three (3) weeks (8)
- 2.2 After six (6) weeks (8)
- 2.3 After nine (9) weeks (8)

QUESTION 3 – RISK ANALYSIS AND EXPECTED MONETARY VALUE**28 MARKS**

- 3.1 Briefly explain what you understand by Expected Monetary Value (EMV) and the role in risk analysis. (4)

- 3.2 In the accompanying scenario, the project manager, Zandile , encountered a risk event when a decision had to be made whether to build or buy a software solution.

The tool that assists in these situations is Expected Monetary Value (EMV).

- a. In the answer sheet provided, complete the EMV calculations based on the scenario presented. (14)
- b. In the answer sheet provided, complete the EMV decision tree based on the scenario presented. Include the calculation needed to complete the decision tree. (8)

QUESTION 4 - COST MANAGEMENT**18 MARKS**

- 4.1 Why is cost containment a major challenge to project managers, and what are some components of project costs that should be included when estimating activity costs? Detail at least two, explaining their relevance in project or activity costing. (6)
- 4.2 What does the term reserve mean and should a reserve amount be included in a project proposal? Explain your answer. (6)
- 4.3 What is the problem with making cost estimates too conservative or too aggressive? (6)

--- END OF PAPER ---

STUDENT NAME: _____ STUDENT No.: _____

APPENDIX – ANSWER SHEETS

Answer to Question 1.1

Earned Value	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9
Design	10	15							
Build									
Testing									
Implementation									
Cumulative									

Answer to Question 2.1 (After 3 weeks)

Indicator	Value	Action Required
CV		
SV		
SPI		
CPI		

Answer to Question 2.2 (After 6 weeks)

Indicator	Value	Action Required
CV		
SV		
SPI		
CPI		

Answer to Question 2.3 (After 9 weeks)

Indicator	Value	Action Required
CV		
SV		
SPI		
CPI		

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APPENDIX – ANSWER SHEETS

Answer to Question 3.2a

RISK EVENT: Decision to make or buy a software component.

Approach 1: Develop software in-house

Options	Probability	Value	Expected Monetary Value
Keep key staff			
Lose key staff			
EMV			

Approach 2: Buy a third-party component

Options	Probability	Value	Expected Monetary Value
Little Customisation			
Extensive Customisation			
EMV			

Answer to Question 3.2b

Decision Tree